

Attack Simulation & Detection Report

Red team exercises against the target VM, with Wazuh detection and MITRE ATT&CK mapping

Author: Edwin Sekgethela Date: May 2026 Status: Complete

1. Build Steps — SOC VM

1.1 Docker Installation

Docker was installed on the SOC VM using the official convenience script, and the working user was added to the Docker group to allow running containers without `sudo` :

```
# Install Docker
curl -fsSL https://get.docker.com | sudo sh

# Allow running containers without sudo
sudo usermod -aG docker <username>
```

1.2 Elasticsearch Memory Configuration

Elasticsearch (used internally by the Wazuh Indexer) requires a minimum `vm.max_map_count` of 262144. This was configured permanently at the OS level:

```
sudo sysctl -w vm.max_map_count=262144
echo "vm.max_map_count=262144" | sudo tee -a /etc/sysctl.conf
```

1.3 Wazuh Stack Deployment

The official Wazuh Docker deployment was cloned and SSL certificates generated before bringing up the full stack:

```
git clone https://github.com/wazuh/wazuh-docker.git -b v4.7.3
cd wazuh-docker/single-node
docker compose -f generate-indexer-certs.yml run --rm generator
docker compose up -d
```

This started three containers: `wazuh.manager` , `wazuh.indexer` , and `wazuh.dashboard` .

Original dashboard credentials redacted for this public version. Default Wazuh installations require the admin password to be changed on first login — do not reuse the Docker Compose default.

2. Build Steps — Target VM

2.1 Wazuh Agent Installation

The Wazuh agent package was downloaded on the SOC VM and transferred to the target VM via SCP, then installed pointing at the SOC VM's manager address:

```
# Download agent package
curl -o /tmp/wazuh-agent.deb https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.3-1_amd64.deb

# Transfer to target VM
scp /tmp/wazuh-agent.deb <user>@192.168.56.20:/tmp/

# Install, pointing at the SOC manager
sudo WAZUH_MANAGER='192.168.56.30' dpkg -i /tmp/wazuh-agent.deb
sudo systemctl enable wazuh-agent && sudo systemctl start wazuh-agent
```

2.2 Attack Surface Configuration

Apache2 was installed to provide an HTTP attack surface on port 80. SSH was already running on port 22 from the base Ubuntu install. Both services were intentionally left running to generate detectable attack traffic.

```
sudo apt install apache2 -y
sudo systemctl enable apache2 && sudo systemctl start apache2
```

SERVICE	PORT	PURPOSE
OpenSSH	22/tcp	SSH brute force target
Apache2	80/tcp	HTTP attack surface
Wazuh Agent	1514/tcp	Log forwarding to SOC manager

3. Attack Simulation & Detection

3.1 Attack 1 — Nmap Reconnaissance

An Nmap service and OS detection scan was run from the Kali VM against the target:

```
nmap -sV -O 192.168.56.20
```

Results revealed:

- Port 22/tcp open — OpenSSH 10.2p1 Ubuntu
- Port 80/tcp open — Apache httpd 2.4.66
- OS detected — Linux (Ubuntu)

SOC DETECTION

Wazuh detected the port scan activity and generated alerts in the Security Events dashboard. The alert timeline showed a spike correlating exactly with the scan timestamp.

3.2 Attack 2 — SSH Brute Force (Hydra)

A dictionary-based SSH brute force attack was launched from the Kali VM using the `rockyou.txt` wordlist:

```
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.20 -t 4 -V
```

Hydra attempted 14,344,399 password combinations against the root account on port 22.

SOC DETECTION

Wazuh detected the brute force attack in real time. Authentication failures spiked from 3 to 113 within seconds. MITRE ATT&CK mapping identified the technique as **T1110 (Brute Force)** under the Credential Access tactic.

4. Alert Summary

Metric	Before Attacks	After Attacks
Total alerts	0	475
Authentication failures	0	113
Authentication successes	14	14
Level 12+ alerts	0	0

4.1 MITRE ATT&CK Techniques Detected

Technique	ID	Tactic	Source
Brute Force	T1110	Credential Access	Hydra SSH attack
Password Guessing	T1110.001	Credential Access	Hydra rockyou.txt
Valid Accounts	T1078	Defence Evasion	Sudo privilege escalation
Remote Services	T1021	Lateral Movement	SSH connections
Sudo and Sudo Caching	T1548.003	Privilege Escalation	Sudo to root

5. Key Findings

- The Wazuh SIEM successfully detected all simulated attacks in real time with zero manual configuration of detection rules.
- SSH brute force attacks generate high-volume authentication failure alerts (rule 5503), immediately visible on the dashboard.
- Wazuh automatically maps detected events to the MITRE ATT&CK framework, providing immediate context for triage.
- The Wazuh agent on the target VM captured authentication events, command execution, file integrity, and system audit events.
- The expanded event view provides full forensic detail: source IP, username, command executed, TTY session, and rule description.

6. Conclusion

This exercise demonstrates a complete security operations workflow — from attack simulation through real-time detection, alert triage, and MITRE ATT&CK mapping. The environment provides a realistic platform for developing SOC analyst skills including SIEM operation, log analysis, threat detection, and incident documentation, built entirely on open-source tools used in production environments globally.

NEXT STEPS

Planned additions: TheHive incident case management, Grafana security dashboards, and additional attack scenarios including web application attacks and lateral movement simulation.

